High Threshold Models

Michael D. Lee

High Threshold Models

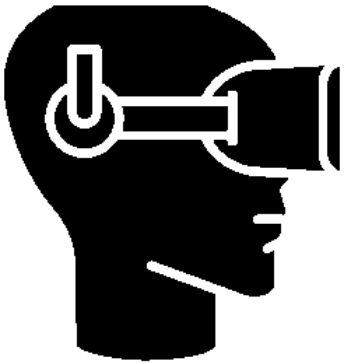
- High threshold models address the same problem as signal detection theory
  - deciding whether or not there is a "signal" or "target" present in the environment

	really signal	really noise
detect signal	hit	false alarm
detect no signal	miss	correct rejection

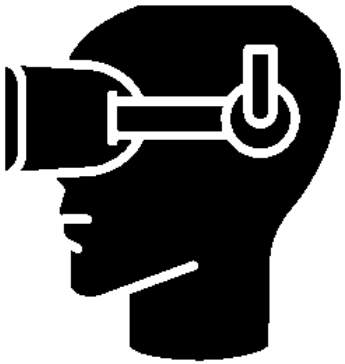
observed behavior

signal detection theory

high threshold

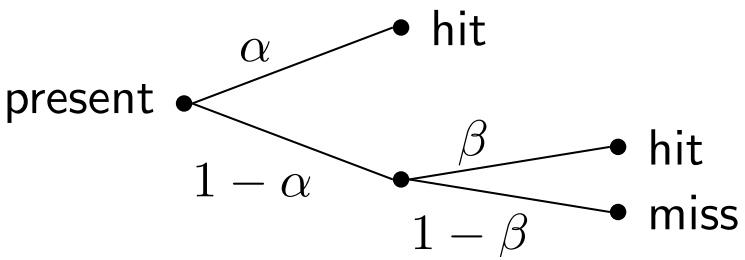


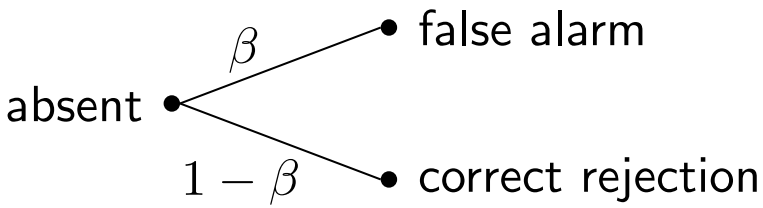




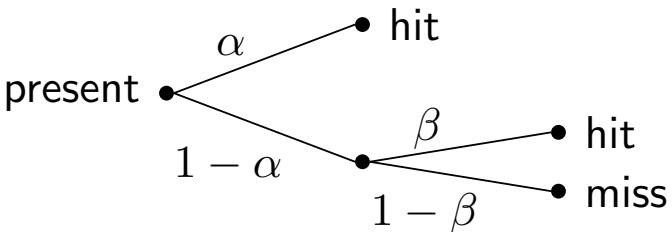
One High and Two High Threshold Models

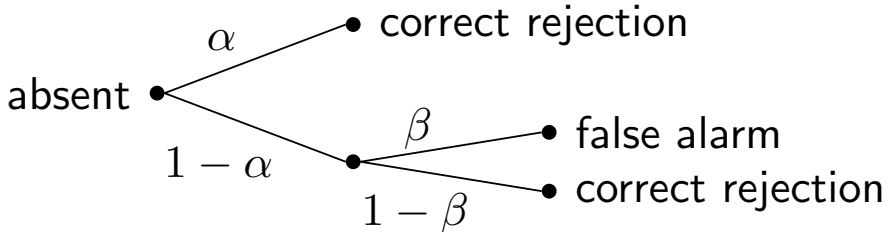
- High Threshold models are examples of a general cognitive modeling framework called Multinomial Processing Trees (Batchelder & Riefer, 1999; Erdfelder et al., 2009)
  - the one high threshold model assumes present signals are detected with probability α
  - otherwise the signal response is guessed with probability β





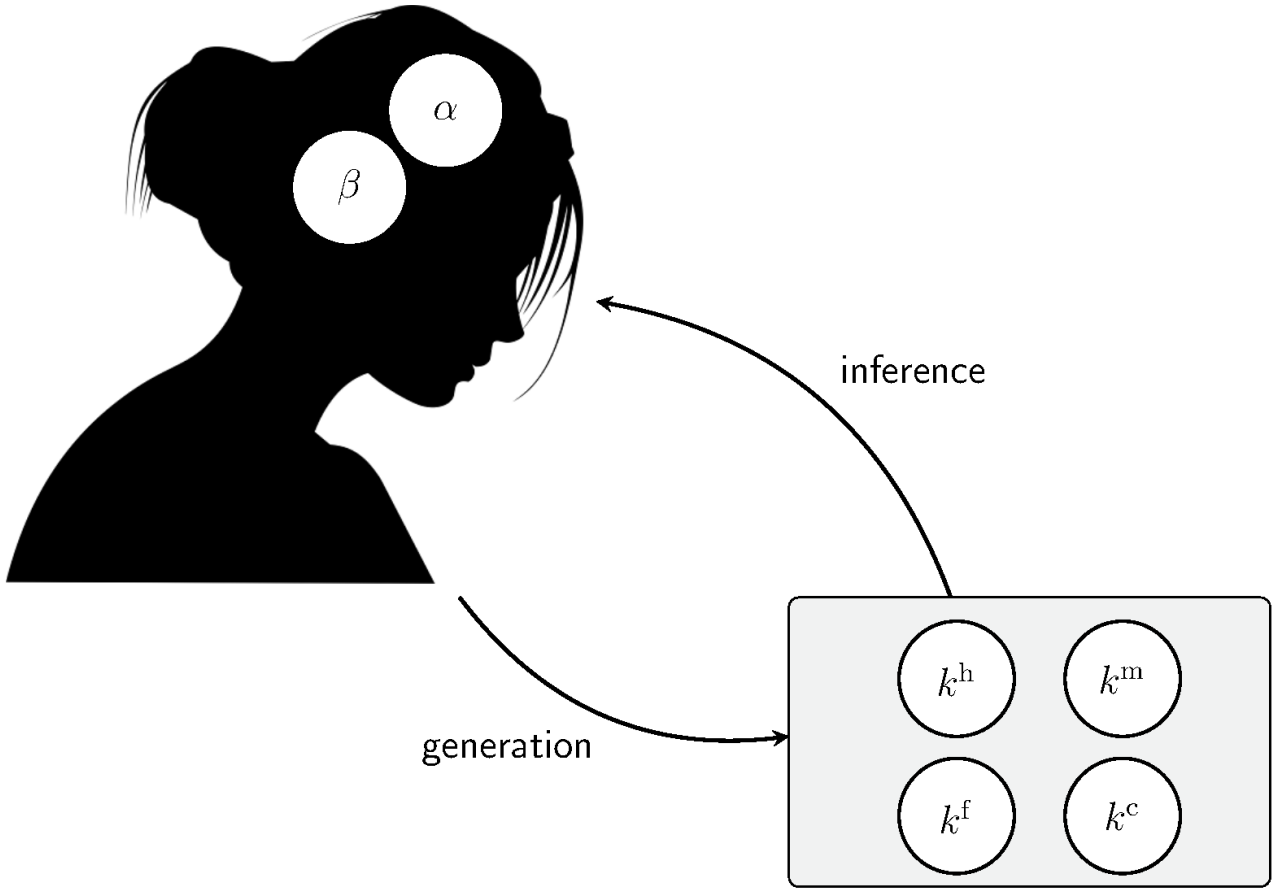
  - the two high threshold model assumes the presence or absence of signals is detected with probability α
  - otherwise the signal response is guessed with probability β





High Threshold Model Structure

- The high threshold models have the structure
  - observed counts of hits k^{h}, false alarms k^{f}, misses k^{m}, and correct rejections k^{c}
  - detection probability α and guessing bias probability β parameters



Head Injury

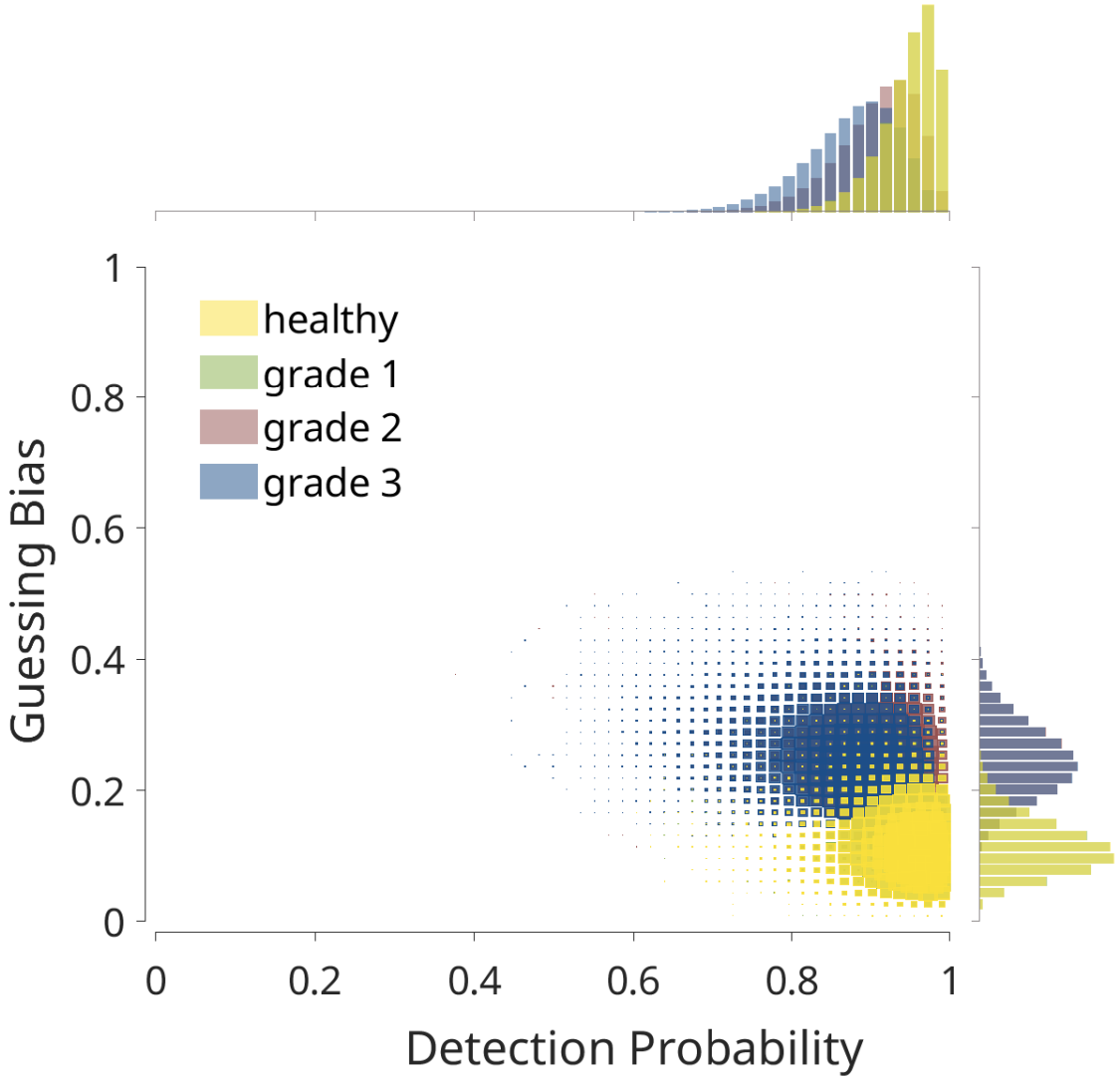
Head Injury Data

- Hannay, Levin, and Grossman (1979) studied changes in recognition memory caused by different grades of concussive head injuries
  - participants was classified as healthy, or one of thre grades of concussion
  - each participant did a 100-trial recognition memory test for drawings
  - 40 old (signal) drawings and 60 new (noise) drawings
- Median individual results in each group are shown below (Hannay et al., 1979, Table II)

	truth							
	healthy		grade 1		grade 2		grade 3	
decision	old	new	old	new	old	new	old	new
old	39	6	39	6	38	14	37	14
new	1	54	1	54	2	46	3	46

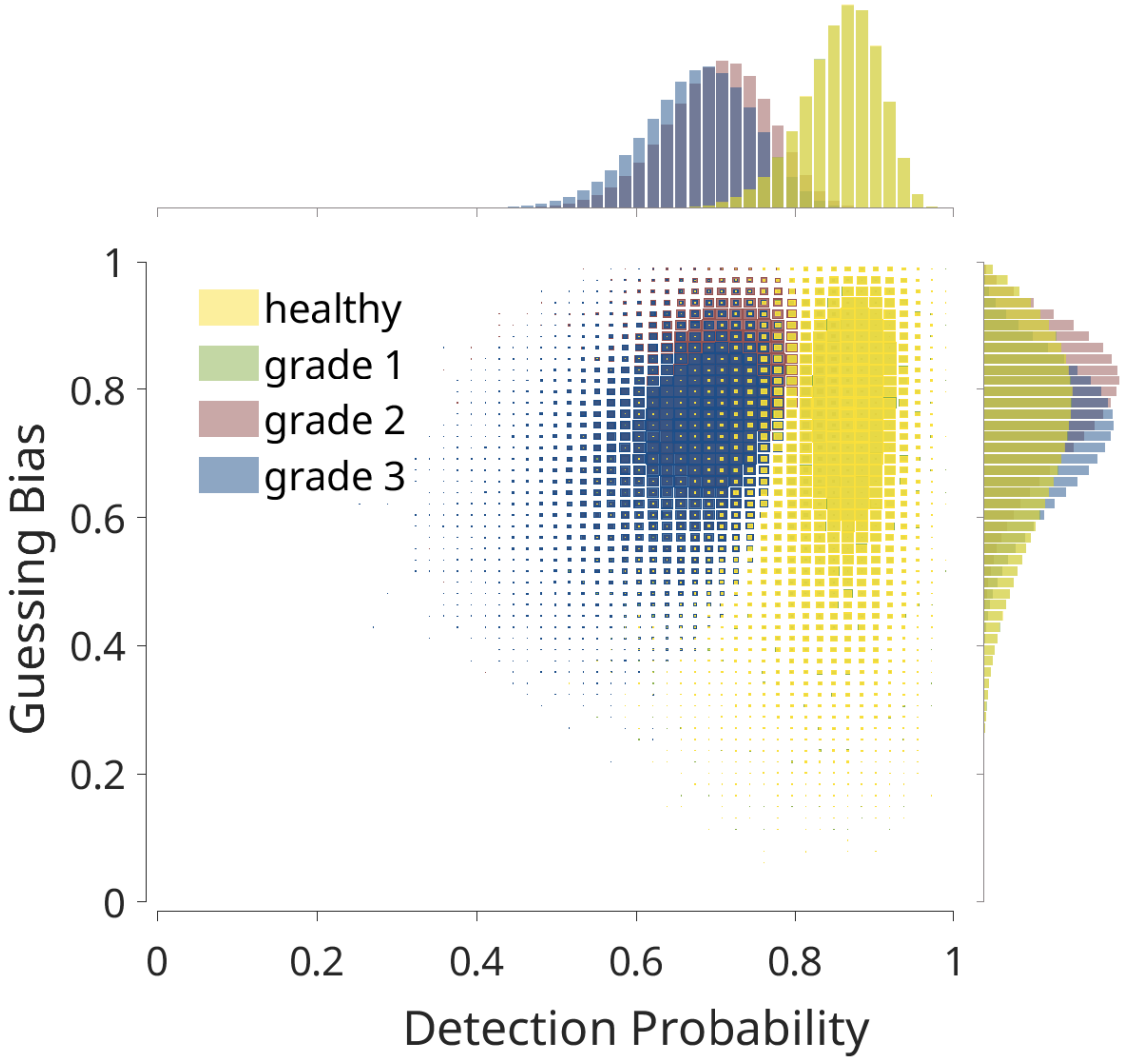
One High Threshold Model Inferences

- Notice that the inference about detection is not entirely determined by the hit rate, because hits can also result from guessing



Two High Threshold Model Inferences

- Detection now is responsible for both hits and correct rejections, and guessing bias applies only if detection has failed in both cases



Citizen Frogs

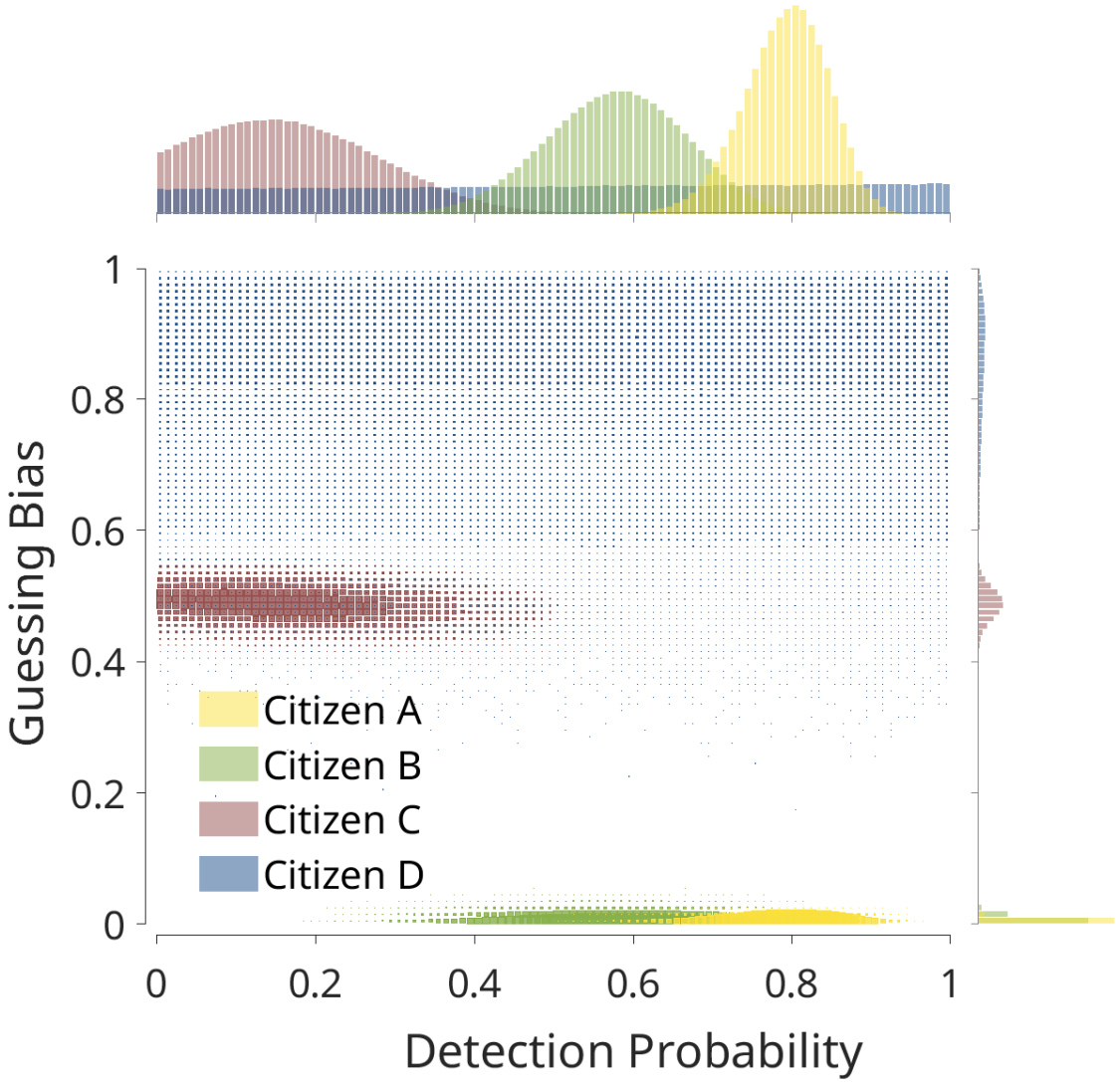
Citizen Frogs Data

- Thorpe et al. (2024) considered data from a "citizen science" project in which non-expert citizens listened to a subset of 1260 audio recordings of frogs
- We consider just 4 of the 587 citizens
  - the three citizens A, B, and C who listened to most recordings
  - one other more typical citizen D who listened to a small number of recordings

	expert ground truth							
	citizen A		citizen B		citizen C		citizen D	
decision	present	absent	present	absent	present	absent	present	absent
"there is a frog"	45	1	17	1	28	227	1	9
"there is no frog"	11	488	12	280	22	237	0	1

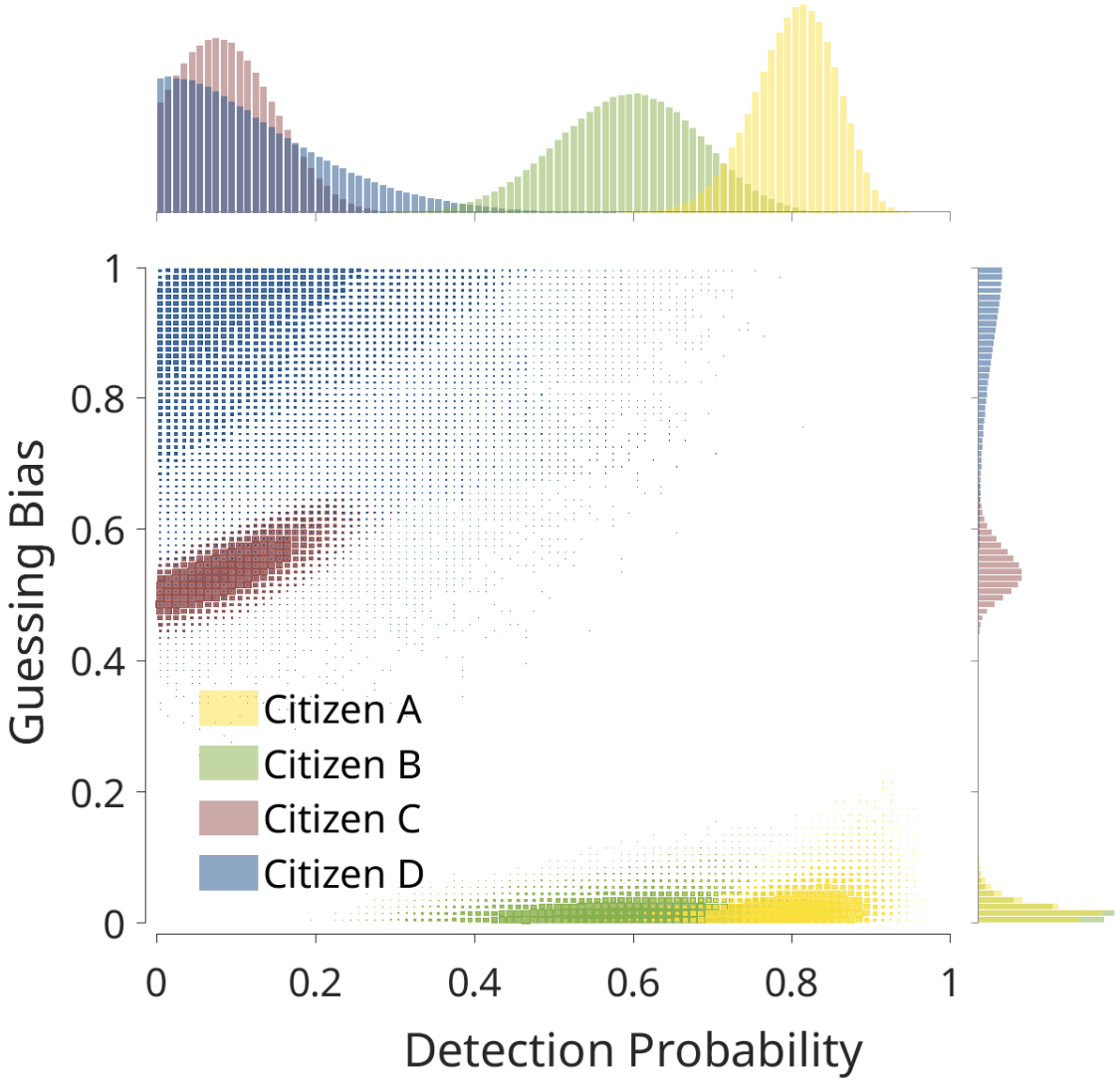
One High Threshold Model Inferences

- Citizen A and B detect well and do not give false alarms, Citizen C is a chance-level detector, and we know little about Citizen D



Two High Threshold Model Inferences

- Citizens A, B, and C have similar inferences, but Citizen D now has poor detection because it has to account for noise trials



Pilot Landing

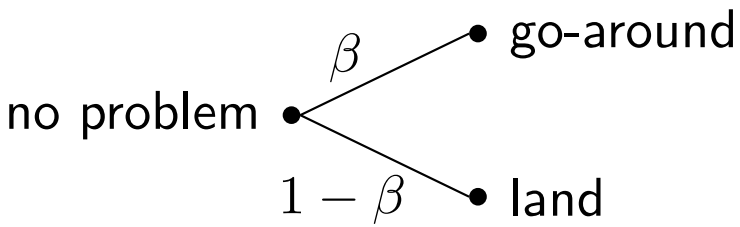
Pilot Landing Data

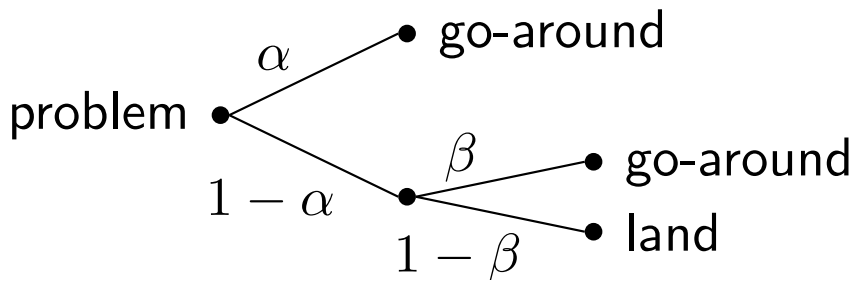
- Salvan, Marois, Chatelais, Lafond, and Audouy (2022) studied the decision making of four pilots in a series of flight simulator trials
  - shown a primary flight display interface with 21 features of the situation (headwind, distance to nearest aircraft, pitch, vertical speed, . . .)
  - had to decide whether to land the plane or execute a "go-around"
- We treat a go-around situation as a signal to be detected

	truth							
	pilot 1		pilot 2		pilot 3		pilot 4	
decision	GA	land	GA	land	GA	land	GA	land
go-around	3	0	2	2	3	3	2	4
land	0	27	1	25	0	24	0	23

One High Threshold Model Interpretation

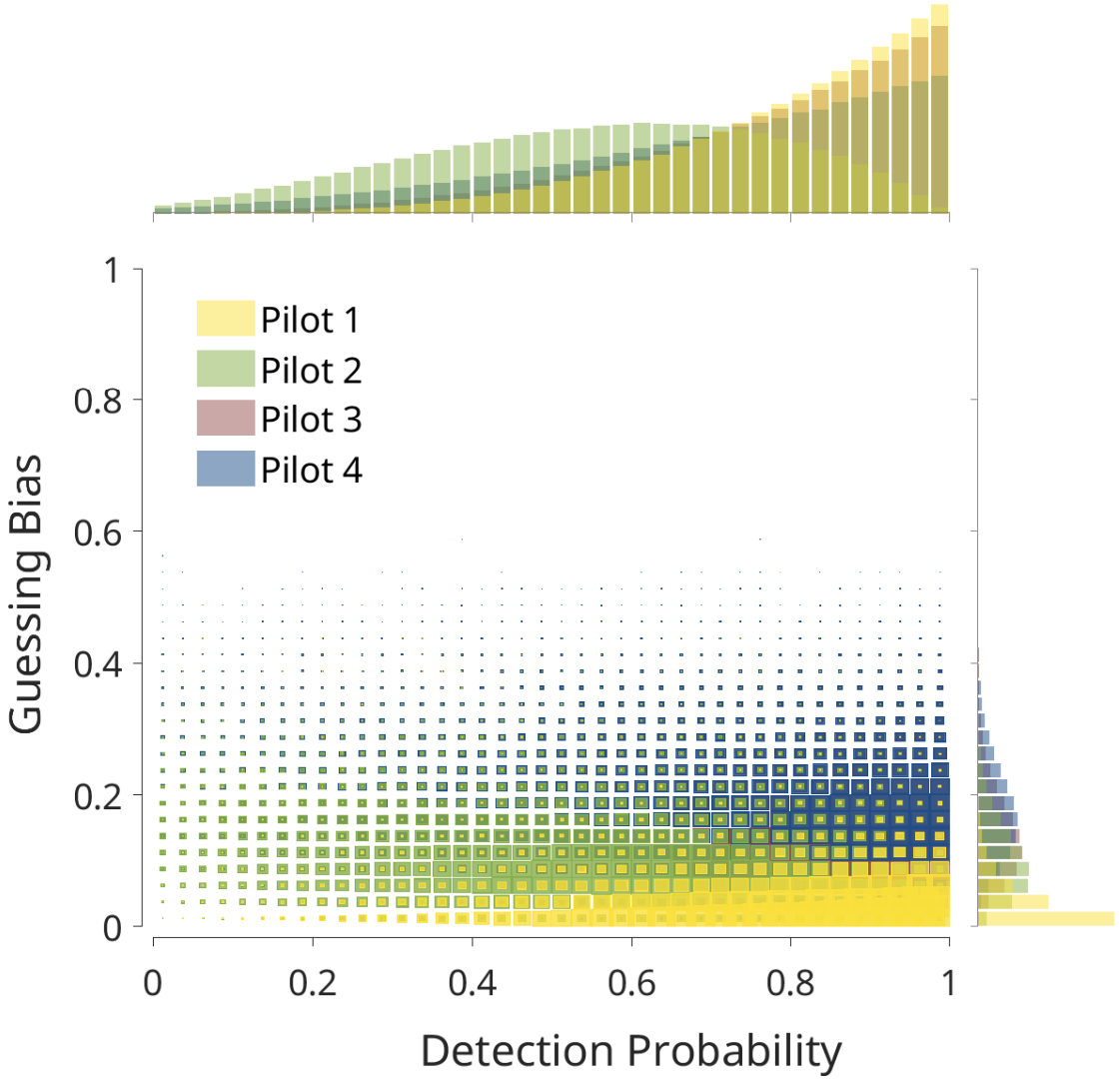
- Each trial either objectively does or does not have a problem
  - in the left tree, there is no problem with landing, but there is still a guessing bias β that the pilot decides to go-around anyway
  - in the right tree, there is a problem, which the pilot notices with detection probability α and executes a go-around
  - in the right tree, if they fail to detect the problem, their behavior is determined by the go-around bias β





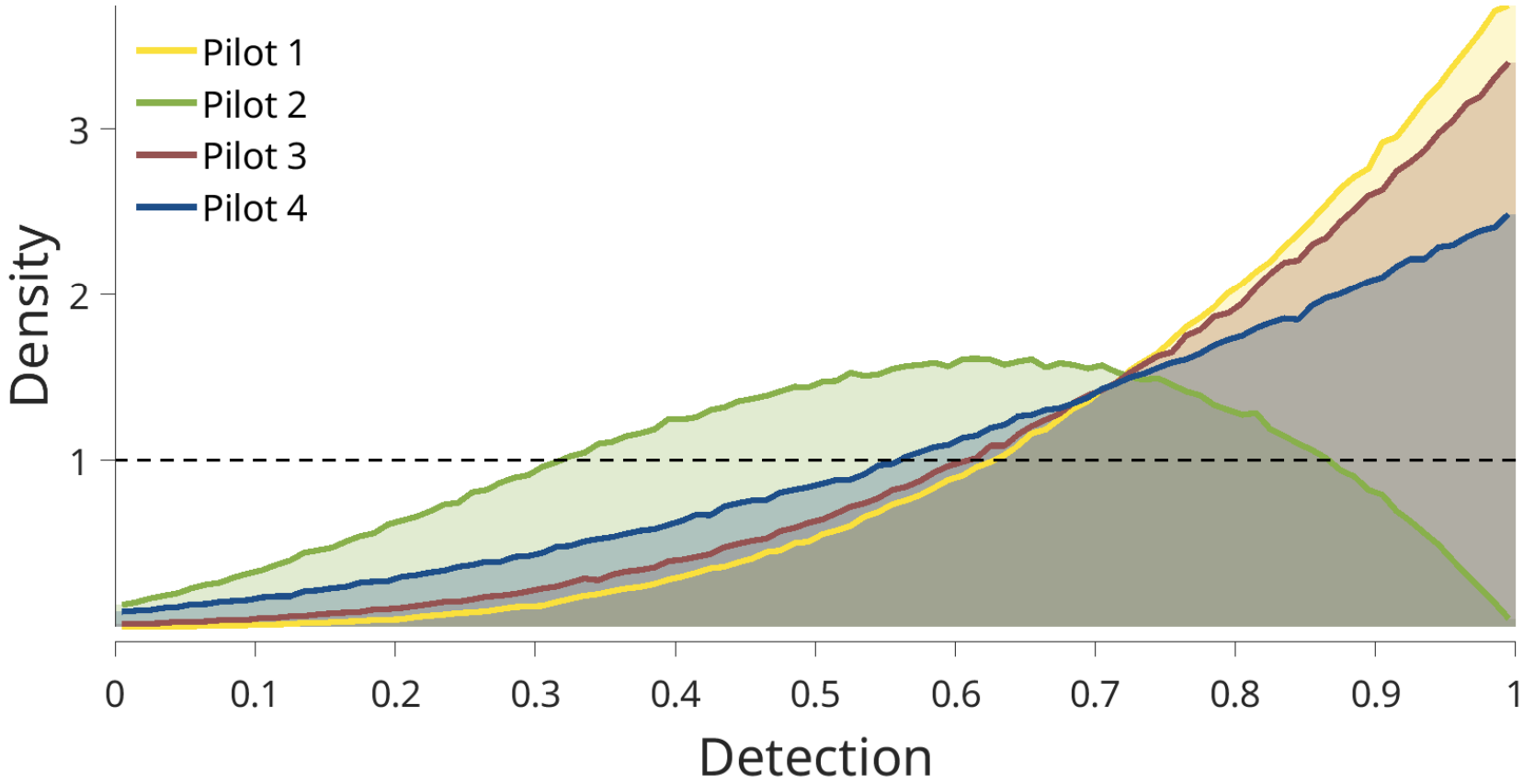
One High Threshold Model Inferences

- The joint posterior distribution looks to be well summarized by the marginal distributions, so we can focus analysis on the marginals



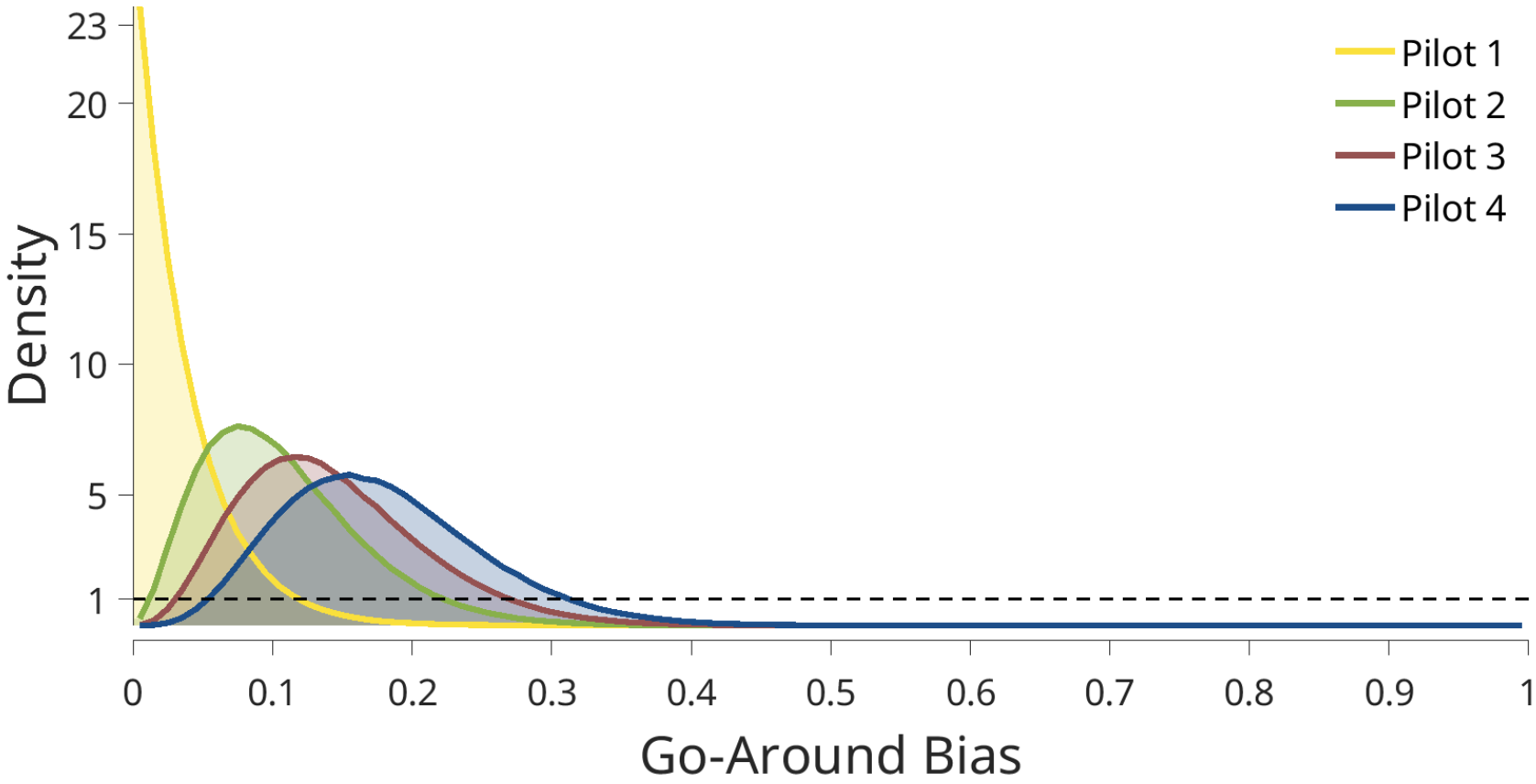
Detection Probability Marginal Inferences

- Pilots 1, 3, and 4 are all likely to have good abilities to detect problems, but pilot 2 is more likely to have a lower ability



Guessing Bias Marginal Inferences

- Pilots 2, 3, and 4 have similar probabilities of going around somewhere between 5% and 30%, but pilot 1 is likely to have a lower bias toward going around



Posterior Predictions

- Another way of understanding the performance of the pilots is provided by posterior predictive analysis
- We assume 1 signal trial and 99 noise trials
  - the motivation is to examine the predicted performance in an expanded set of 100 trials that still follow a real-world base rate with 1% of landings involving a problem that needs to be detected
- The mean and 95% credible intervals the posterior predicted hit and false alarm counts for each pilot in this situation are shown below

	hits		false alarms	
	mean	95% CI	mean	95% CI
pilot 1	0.80	$(0, 1)$	3.53	$(0, 14)$
pilot 2	0.60	$(0, 1)$	10.54	$(1, 26)$
pilot 3	0.80	$(0, 1)$	14.14	$(3, 31)$
pilot 4	0.76	$(0, 1)$	17.63	$(5, 35)$

References i

Batchelder, W. H., & Riefer, D. M. (1999). Theoretical and empirical review of multinomial process tree modeling. *Psychonomic Bulletin & Review*, *6*, 57–86.

Erdfelder, E., Auer, T.-S., Hilbig, B. E., Aßfalg, A., Moshagen, M., & Nadarevic, L. (2009). Multinomial processing tree models: A review of the literature. *Zeitschrift für Psychologie/Journal of Psychology*, *217*, 108–124.

Hannay, H. J., Levin, H. S., & Grossman, R. G. (1979). Impaired recognition memory after head injury. *Cortex*, *15*(2), 269–283.

Salvan, L., Marois, A., Chatelais, B., Lafond, D., & Audouy, A. (2022). Pilots' decision-making during unstabilized approaches: Group-level policy capturing for cognitive assistance. In *International conference on cognitive aircraft systems.* Retrieved from https://events.isae-supaero.fr/event/14/contributions/388/attachments/53/91/ICCAS2022_Salvan_et_al_extended_abstract.pdf

References ii

Thorpe, A., Kelly, O., Callen, A., Griffin, A. S., & Brown, S. D. (2024). Using a cognitive model to understand crowdsourced data from citizen scientists. *Behavior Research Methods*, *56*(4), 3589–3605. doi: https://doi.org/10.3758/s13428-023-02289-w